

30 EJERCICIOS PRÁCTICOS DEL MÓDULO 5

Divididos en básicos, intermedios, avanzados y proyecto final.

✓ EJERCICIOS BÁSICOS (1–10)

1. Mostrar todos los procesos activos

ps aux

```
jcsantos@JCSantos-Ubuntu:~/Ejercicios_Linux/aula
jcsantos at JCSantos-Ubuntu in ~/Ejercicios_Linux/aula
ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.2 26072 16600 ?        Ss   09:01   0:09 /usr/lib/systemd/systemd --switched-root --system
root         2  0.0  0.0      0     0 ?        S    09:01   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    09:01   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0     0 ?        I<   09:01   0:00 [kworker/R-rcu_gp]
root         5  0.0  0.0      0     0 ?        I<   09:01   0:00 [kworker/R-sync_wq]
```

2. Mostrar procesos en formato árbol

ps tree

```
jcsantos@JCSantos-Ubuntu:~/Ejercicios_Linux/aula
jcsantos at JCSantos-Ubuntu in ~/Ejercicios_Linux/aula
ps tree
systemd--ModemManager--3*[{ModemManager}]
         --NetworkManager--3*[{NetworkManager}]
         --VGAuthService
         --accounts-daemon--3*[{accounts-daemon}]
         --avahi-daemon--avahi-daemon
         --chronyd-starter--chronyd--chronyd
```

3. Ver los 10 procesos que más CPU utilizan

top

Usa P dentro de top.

```
top - top
~/Ejercicios_Linux/aula

top - 11:53:46 up 2:52, 1 user, load average: 0.10, 0.08, 0.02
Tasks: 367 total, 1 running, 361 sleeping, 0 stopped, 5 zombie
%Cpu(s): 0.1 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7381.1 total, 4010.5 free, 1575.3 used, 2112.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 5805.9 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
11561 jcsantos  20   0   22232   6272   4052  R   0.7   0.1   0:00.06 top
2372  jcsantos  20   0   742728 104272 83864  S   0.3   1.4   0:21.08 vmttoolsd
3242  jcsantos  20   0 1684800 237000 112016 S   0.3   3.1   0:56.79 ptyxis
   1 root      20   0   26072   16600 11052  S   0.0   0.2   0:09.93 systemd
   2 root      20   0      0      0      0  S   0.0   0.0   0:00.02 kthreadd
   3 root      20   0      0      0      0  S   0.0   0.0   0:00.00 pool_workqueue_release
   4 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-rcu_gp
   5 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-sync_wq
   6 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
   7 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
   8 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/R-netns
  11 root      0 -20      0      0      0  I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
  12 root      20   0      0      0      0  I   0.0   0.0   0:00.00 kworker/u512:0-ipv6_addrconf
```

4. Ver los procesos que más memoria consumen

top

Usa M.

```
top - 11:54:36 up 2:53, 1 user, load average: 0.08, 0.08, 0.02
Tasks: 367 total, 1 running, 361 sleeping, 0 stopped, 5 zombie
%Cpu(s): 0.7 us, 0.4 sy, 0.0 ni, 98.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7381.1 total, 4009.9 free, 1575.9 used, 2112.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 5805.3 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 2143 jcsantos  20   0  4483960 295564 136700 S   5.6   3.9   1:43.55 gnome-shell
 3242 jcsantos  20   0 1684748 237000 112016 S   2.0   3.1   0:57.33 ptyxis
 2372 jcsantos  20   0  742880 104424  83868 S   0.3   1.4   0:21.19 vmtoolsd
 2811 jcsantos  20   0  757096 102212  80296 S   0.0   1.4   0:00.29 mutter-x11-fram
 2691 jcsantos  20   0  253104  96672  64528 S   0.0   1.3   0:00.95 Xwayland
 2862 jcsantos  20   0  2935472  65720  51024 S   0.0   0.9   0:01.14 gjs
 2365 jcsantos  20   0  916968  64824  51392 S   0.0   0.9   0:00.23 evolution-alarm
 4081 root      20   0  498284  43908  37508 S   0.0   0.6   0:01.52 fwupd
 2294 jcsantos  20   0 1028856  43724  35536 S   0.0   0.6   0:00.18 evolution-sourc
 2853 jcsantos  20   0  549908  42908  32856 S   0.0   0.6   0:00.43 xdg-desktop-por
  543 root      19  -1   73036  42384  40704 S   0.0   0.6   0:00.83 systemd-journal
 1356 root      20   0 1745284  41776  26504 S   0.0   0.6   0:01.71 snapd
 2557 jcsantos  20   0  507736  34896  20940 S   0.0   0.5   0:03.39 ibus-extension-
 2373 jcsantos  20   0  578004  32496  25132 S   0.0   0.4   0:00.51 update-notifier
```

5. Filtrar procesos por usuario

ps -u \$USER

```
jcsantos@JCSantos-Ubunut:~/Ejercicios_Linux/aula
ps -u $USER
  PID TTY          TIME CMD
 1950 ?        00:00:05 systemd
 1964 ?        00:00:00 (sd-pam)
 1993 ?        00:00:00 dbus-daemon
 1994 ?        00:02:01 pipewire
 1995 ?        00:00:00 pipewire
 1999 ?        00:00:00 gnome-keyring-d
 2015 ?        00:00:00 mpris-proxy
 2016 ?        00:00:02 wireplumber
 2019 ?        00:00:00 pipewire-pulse
```

6. Encontrar el PID de un programa específico

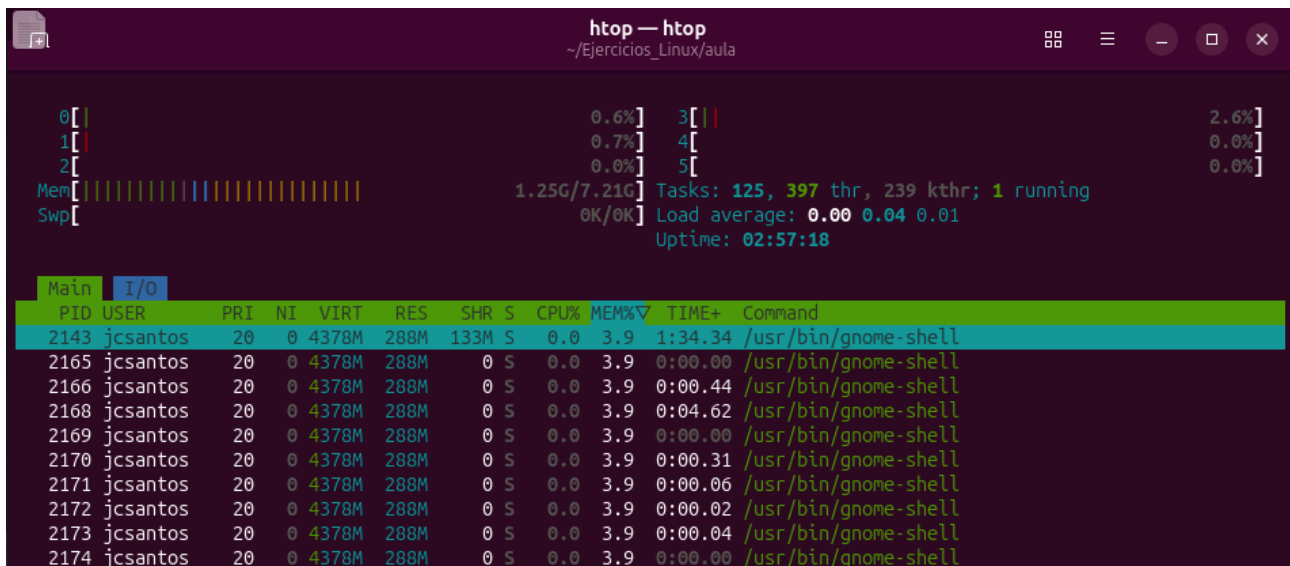
pidof systemd

pidof bash

```
jcsantos@JCSantos-Ubunut:~/Ejercicios_Linux/aula
pidof systemd
1950 1
jcsantos@JCSantos-Ubunut:~/Ejercicios_Linux/aula
pidof bash
7790 3284
jcsantos@JCSantos-Ubunut:~/Ejercicios_Linux/aula
pidof zsh
jcsantos@JCSantos-Ubunut:~/Ejercicios_Linux/aula
```

7. Abrir htop y ordenar por RAM

Pulsa F6.



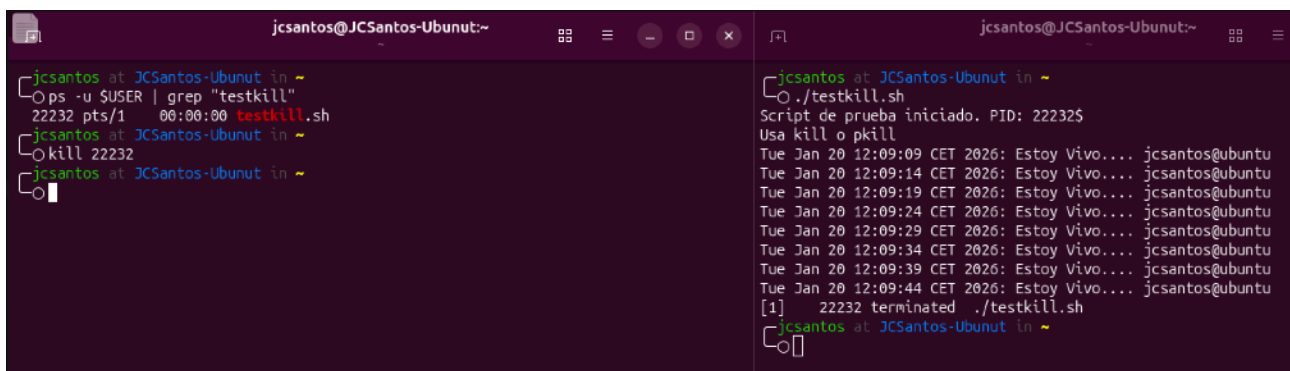
```
htop — htop
~/Ejercicios_Linux/aula

0[ | 0.6%] 3[ | 2.6%]
1[ | 0.7%] 4[ | 0.0%]
2[ | 0.0%] 5[ | 0.0%]
Mem[|||||] 1.25G/7.21G Tasks: 125, 397 thr, 239 kthr; 1 running
Swp[ ] 0K/0K Load average: 0.00 0.04 0.01
Uptime: 02:57:18

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
2143 jcsantos 20 0 4378M 288M 133M S 0.0 3.9 1:34.34 /usr/bin/gnome-shell
2165 jcsantos 20 0 4378M 288M 0 S 0.0 3.9 0:00.00 /usr/bin/gnome-shell
2166 jcsantos 20 0 4378M 288M 0 S 0.0 3.9 0:00.44 /usr/bin/gnome-shell
2168 jcsantos 20 0 4378M 288M 0 S 0.0 3.9 0:04.62 /usr/bin/gnome-shell
2169 jcsantos 20 0 4378M 288M 0 S 0.0 3.9 0:00.00 /usr/bin/gnome-shell
2170 jcsantos 20 0 4378M 288M 0 S 0.0 3.9 0:00.31 /usr/bin/gnome-shell
2171 jcsantos 20 0 4378M 288M 0 S 0.0 3.9 0:00.06 /usr/bin/gnome-shell
2172 jcsantos 20 0 4378M 288M 0 S 0.0 3.9 0:00.02 /usr/bin/gnome-shell
2173 jcsantos 20 0 4378M 288M 0 S 0.0 3.9 0:00.04 /usr/bin/gnome-shell
2174 jcsantos 20 0 4378M 288M 0 S 0.0 3.9 0:00.00 /usr/bin/gnome-shell
```

8. Matar un proceso por PID

kill PID

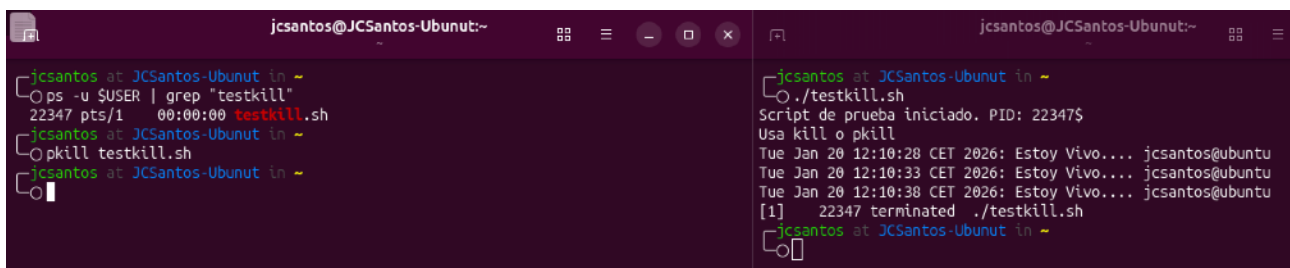


```
jcsantos@JCSantos-Ubunut:~
jcsantos at JCSantos-Ubunut in ~
ps -u $USER | grep "testkill"
22232 pts/1 00:00:00 testkill.sh
jcsantos at JCSantos-Ubunut in ~
kill 22232
jcsantos at JCSantos-Ubunut in ~

jcsantos@JCSantos-Ubunut:~
jcsantos at JCSantos-Ubunut in ~
./testkill.sh
Script de prueba iniciado. PID: 222325
Usa kill o pkill
Tue Jan 20 12:09:09 CET 2026: Estoy Vivo.... jcsantos@ubuntu
Tue Jan 20 12:09:14 CET 2026: Estoy Vivo.... jcsantos@ubuntu
Tue Jan 20 12:09:19 CET 2026: Estoy Vivo.... jcsantos@ubuntu
Tue Jan 20 12:09:24 CET 2026: Estoy Vivo.... jcsantos@ubuntu
Tue Jan 20 12:09:29 CET 2026: Estoy Vivo.... jcsantos@ubuntu
Tue Jan 20 12:09:34 CET 2026: Estoy Vivo.... jcsantos@ubuntu
Tue Jan 20 12:09:39 CET 2026: Estoy Vivo.... jcsantos@ubuntu
Tue Jan 20 12:09:44 CET 2026: Estoy Vivo.... jcsantos@ubuntu
[1] 22232 terminated ./testkill.sh
jcsantos at JCSantos-Ubunut in ~
```

9. Matar procesos por nombre

pkill firefox



```
jcsantos@JCSantos-Ubunut:~
jcsantos at JCSantos-Ubunut in ~
ps -u $USER | grep "testkill"
22347 pts/1 00:00:00 testkill.sh
jcsantos at JCSantos-Ubunut in ~
pkill testkill.sh
jcsantos at JCSantos-Ubunut in ~

jcsantos@JCSantos-Ubunut:~
jcsantos at JCSantos-Ubunut in ~
./testkill.sh
Script de prueba iniciado. PID: 22347$
Usa kill o pkill
Tue Jan 20 12:10:28 CET 2026: Estoy Vivo.... jcsantos@ubuntu
Tue Jan 20 12:10:33 CET 2026: Estoy Vivo.... jcsantos@ubuntu
Tue Jan 20 12:10:38 CET 2026: Estoy Vivo.... jcsantos@ubuntu
[1] 22347 terminated ./testkill.sh
jcsantos at JCSantos-Ubunut in ~
```

10. Ejecutar un comando con prioridad modificada

`nice -n 5 sleep 200`



The screenshot shows two terminal windows. The left window, titled 'htop - htop', displays system statistics and a table of running processes. The right window, titled 'nice -n 100 ./testkill.sh', shows the execution of a script that uses 'nice' to run a process with a lower priority.

```
htop - htop
0[ 1.3%] 3[ 0.7%]
1[ 0.0%] 4[ 0.0%]
2[ 0.0%] 5[ 0.0%]
Mem[|||||] 1.22G/7.21G Tasks: 127, 397 thr, 237 kthr; 1 running
Swp[ 0K/0K] Load average: 0.08 0.05 0.03
Uptime: 03:12:50

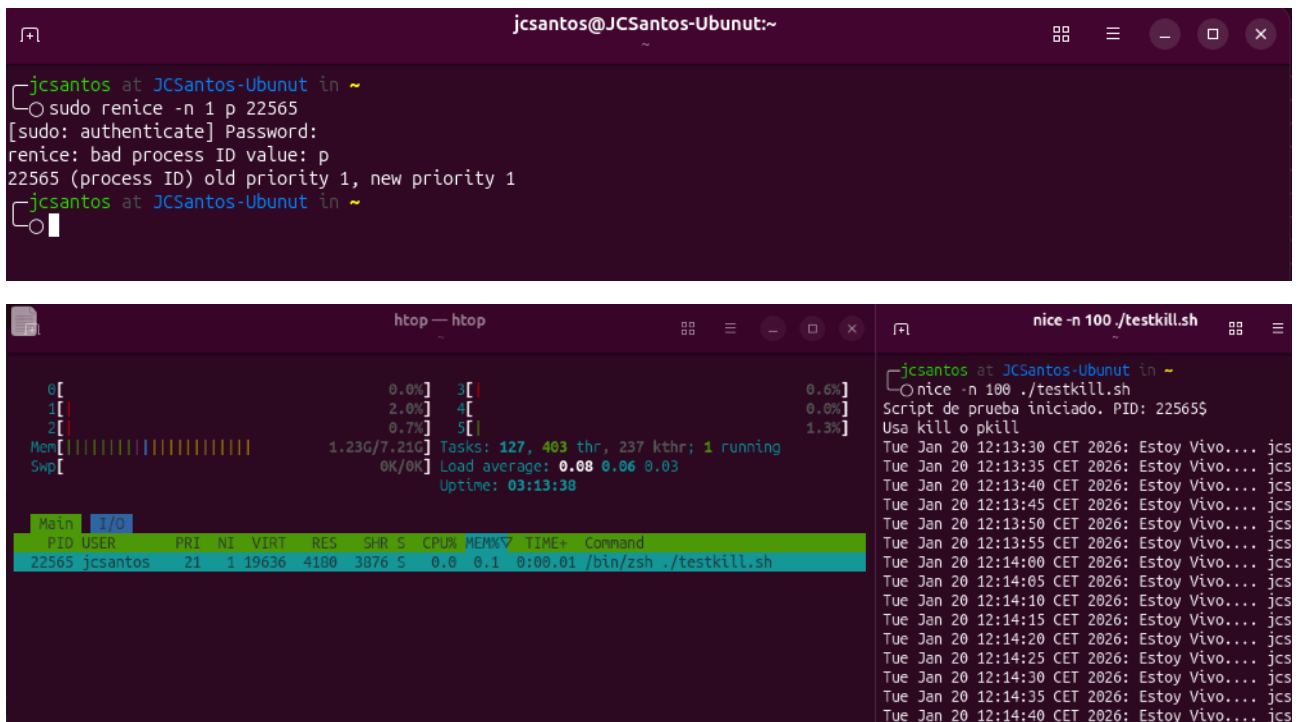
Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM%+ TIME+ Command
22565 jcsantos 39 19 19636 4188 3876 S 0.0 0.1 0:00.00 /bin/zsh ./testkill.sh

nice -n 100 ./testkill.sh
jcsantos at JCSantos-Ubunut in ~
$ nice -n 100 ./testkill.sh
Script de prueba iniciado. PID: 22565$
Usa kill o pkill
Tue Jan 20 12:13:30 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:13:35 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:13:40 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:13:45 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:13:50 CET 2026: Estoy Vivo.... jcs
```

✓ EJERCICIOS INTERMEDIOS (11-20)

11. Cambiar la prioridad de un proceso en ejecución

`sudo renice -n 10 -p PID`



The screenshot shows two terminal windows. The left window, titled 'jcsantos@JCSantos-Ubunut:~', shows the execution of 'sudo renice -n 10 -p 22565'. The right window, titled 'nice -n 100 ./testkill.sh', shows the execution of a script that uses 'nice' to run a process with a lower priority.

```
jcsantos@JCSantos-Ubunut:~
$ sudo renice -n 10 -p 22565
[sudo: authenticate] Password:
renice: bad process ID value: p
22565 (process ID) old priority 1, new priority 1
jcsantos at JCSantos-Ubunut in ~
$

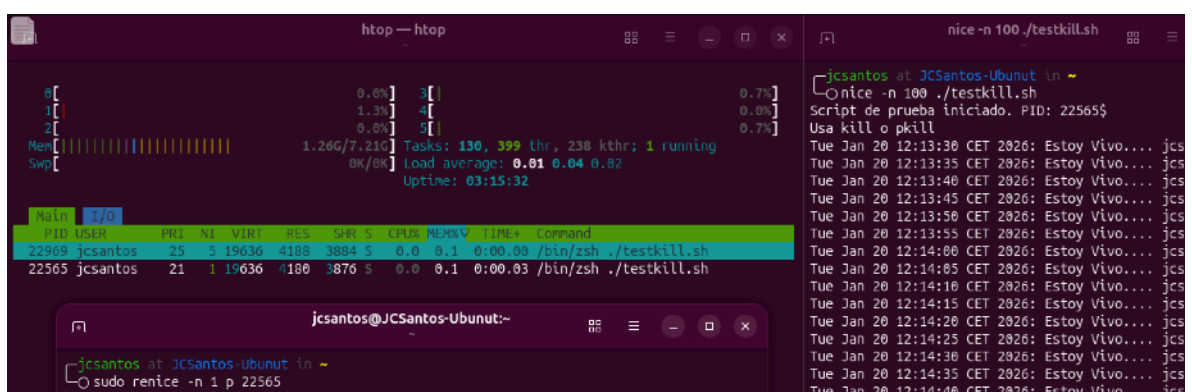
htop - htop
0[ 0.0%] 3[ 0.6%]
1[ 2.0%] 4[ 0.0%]
2[ 0.7%] 5[ 1.3%]
Mem[|||||] 1.23G/7.21G Tasks: 127, 403 thr, 237 kthr; 1 running
Swp[ 0K/0K] Load average: 0.08 0.06 0.03
Uptime: 03:13:38

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM%+ TIME+ Command
22565 jcsantos 21 1 19636 4188 3876 S 0.0 0.1 0:00.01 /bin/zsh ./testkill.sh

nice -n 100 ./testkill.sh
jcsantos at JCSantos-Ubunut in ~
$ nice -n 100 ./testkill.sh
Script de prueba iniciado. PID: 22565$
Usa kill o pkill
Tue Jan 20 12:13:30 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:13:35 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:13:40 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:13:45 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:13:50 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:13:55 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:14:00 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:14:05 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:14:10 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:14:15 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:14:20 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:14:25 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:14:30 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:14:35 CET 2026: Estoy Vivo.... jcs
Tue Jan 20 12:14:40 CET 2026: Estoy Vivo.... jcs
```

12. Crear proceso en segundo plano

`sleep 300 &`



The screenshot shows two terminal windows. The left window, titled 'htop - htop', displays system statistics and a table of running processes. The right window, titled 'nice -n 100 ./testkill.sh', shows the execution of a script that uses 'nice' to run a process with a lower priority. A third terminal window is visible in the foreground, showing the execution of 'sleep 300 &'.

```
htop - htop
0[ 0.0%] 3[ 0.7%]
1[ 1.3%] 4[ 0.0%]
2[ 0.0%] 5[ 0.7%]
Mem[|||||] 1.26G/7.21G Tasks: 130, 399 thr, 238 kthr; 1 running
Swp[ 0K/0K] Load average: 0.01 0.04 0.02
Uptime: 03:15:32

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM%+ TIME+ Command
22565 jcsantos 21 1 19636 4188 3876 S 0.0 0.1 0:00.03 /bin/zsh ./testkill.sh

jcsantos@JCSantos-Ubunut:~
$ sleep 300 &
[1] 22566
jcsantos at JCSantos-Ubunut in ~
$
```

13. Traer proceso al primer plano

fg

```
jcsantos at JCSantos-Ubunut in ~  
└─○ fg  
[1] + 22969 running ./testkill.sh  
Tue Jan 20 12:17:06 CET 2026: Estoy Vivo.... jcsantos@ubuntu  
Tue Jan 20 12:17:11 CET 2026: Estoy Vivo.... jcsantos@ubuntu  
Tue Jan 20 12:17:16 CET 2026: Estoy Vivo.... jcsantos@ubuntu
```

14. Enviar proceso al segundo plano

Ctrl + Z

bg

15. Ver trabajos en background

jobs

```
└─○ ./testkill.sh &  
[1] 31069  
Script de prueba iniciado. PID: 31069$  
Usa kill o pkill  
Tue Jan 20 14:02:15 CET 2026: Estoy Vivo.... jcsantos@ubuntu  
jcsantos at JCSantos-Ubunut in ~  
└─○ jobs  
[1] + running ./testkill.sh  
jcsantos at JCSantos-Ubunut in ~  
└─○ Tue Jan 20 14:02:20 CET 2026: Estoy Vivo.... jcsantos@ubuntu  
Tue Jan 20 14:02:25 CET 2026: Estoy Vivo.... jcsantos@ubuntu
```

16. Ver servicios activos

`systemctl --type=service --state=running`

```
systemctl --type=service --state=running — systemctl --type=ser...
jcsantos at JCSantos-Ubunut in ~
○ systemctl --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
accounts-daemon.service            loaded active running Accounts Service
avahi-daemon.service               loaded active running Avahi mDNS/DNS-SD Stack
chrony.service                     loaded active running chrony, an NTP client/server
colord.service                     loaded active running Manage, Install and Generate Color
cron.service                       loaded active running Regular background program process
cups-browsed.service               loaded active running Make remote CUPS printers availabl
cups.service                       loaded active running CUPS Scheduler
dbus.service                       loaded active running D-Bus System Message Bus
fwupd.service                      loaded active running Firmware update daemon
gdm.service                        loaded active running GNOME Display Manager
ModemManager.service              loaded active running Modem Manager
networkd-dispatcher.service        loaded active running Dispatcher daemon for systemd-netw
NetworkManager.service            loaded active running Network Manager
open-vm-tools.service              loaded active running Service for virtual machines hoste
polkit.service                     loaded active running Authorization Manager
power-profiles-daemon.service       loaded active running Power Profiles daemon
rsyslog.service                   loaded active running System Logging Service
rtkit-daemon.service              loaded active running RealtimeKit Scheduling Policy Serv
snapd.service                      loaded active running Snap Daemon
switcheroo-control.service         loaded active running Switcheroo Control Proxy service
systemd-journald.service           loaded active running Journal Service
systemd-logind.service             loaded active running User Login Management
systemd-oomd.service              loaded active running Userspace Out-Of-Memory (OOM) Kill
systemd-resolved.service           loaded active running Network Name Resolution
systemd-udevd.service              loaded active running Rule-based Manager for Device Even
udisks2.service                   loaded active running Disk Manager
unattended-upgrades.service        loaded active running Unattended Upgrades Shutdown
upower.service                     loaded active running Daemon for power management
user@1000.service                 loaded active running User Manager for UID 1000
vgauth.service                     loaded active running Authentication service for virtual
wpa_supplicant.service             loaded active running WPA supplicant

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
         ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
         SUB    → The low-level unit activation state, values depend on unit type.
```

17. Mostrar estado del servicio ssh

`systemctl status ssh`

```
jcsantos at JCSantos-Ubunut in ~
○ systemctl status ssh
Unit ssh.service could not be found.
jcsantos at JCSantos-Ubunut in ~
○ sudo apt install ssh
[sudo: authenticate] Password:
Installing:
  ssh

Installing dependencies:
  ncurses-term  openssh-server  openssh-sftp-server  ssh-import-id
```

18. Detener el servicio ssh

sudo systemctl stop ssh

19. Iniciar el servicio ssh

sudo systemctl start ssh

```
jcsantos at JCSantos-Ubunut in ~
└─○ sudo systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead) since Tue 2026-01-20 14:16:34 CET; 1min 3s ago
   Duration: 7.695s
   Invocation: 9a4e8ec3c2b14a40850b21a868920f71
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 2868 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Process: 2870 ExecStart=/usr/sbin/sshd -D $SSHD_OPTS (code=exited, status=0/SUCCESS)
   Main PID: 2870 (code=exited, status=0/SUCCESS)
   Mem peak: 2.4M
   CPU: 25ms

Jan 20 14:16:26 JCSantos-Ubunut systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jan 20 14:16:26 JCSantos-Ubunut sshd[2870]: Server listening on 0.0.0.0 port 22.
Jan 20 14:16:26 JCSantos-Ubunut sshd[2870]: Server listening on :: port 22.
Jan 20 14:16:26 JCSantos-Ubunut systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Jan 20 14:16:34 JCSantos-Ubunut sshd[2870]: Received signal 15; terminating.
Jan 20 14:16:34 JCSantos-Ubunut systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
Jan 20 14:16:34 JCSantos-Ubunut systemd[1]: ssh.service: Deactivated successfully.
Jan 20 14:16:34 JCSantos-Ubunut systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.

jcsantos at JCSantos-Ubunut in ~
└─○ sudo systemctl stop ssh.socket
jcsantos at JCSantos-Ubunut in ~
└─○ sudo systemctl stop ssh.service
jcsantos at JCSantos-Ubunut in ~
└─○ sudo systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead) since Tue 2026-01-20 14:16:34 CET; 1min 31s ago
   Duration: 7.695s
   Invocation: 9a4e8ec3c2b14a40850b21a868920f71
   TriggeredBy: ○ ssh.socket
   Docs: man:sshd(8)
```

20. Ver logs del servicio ssh

journalctl -u ssh

```
jcsantos at JCSantos-Ubunut in ~
└─○ journalctl -u ssh
Jan 20 14:16:26 JCSantos-Ubunut systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jan 20 14:16:26 JCSantos-Ubunut sshd[2870]: Server listening on 0.0.0.0 port 22.
Jan 20 14:16:26 JCSantos-Ubunut sshd[2870]: Server listening on :: port 22.
Jan 20 14:16:26 JCSantos-Ubunut systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Jan 20 14:16:34 JCSantos-Ubunut sshd[2870]: Received signal 15; terminating.
Jan 20 14:16:34 JCSantos-Ubunut systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
Jan 20 14:16:34 JCSantos-Ubunut systemd[1]: ssh.service: Deactivated successfully.
Jan 20 14:16:34 JCSantos-Ubunut systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
Jan 20 14:19:13 JCSantos-Ubunut systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jan 20 14:19:13 JCSantos-Ubunut sshd[3106]: Server listening on 0.0.0.0 port 22.
Jan 20 14:19:13 JCSantos-Ubunut sshd[3106]: Server listening on :: port 22.
Jan 20 14:19:13 JCSantos-Ubunut systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
└─○
```

✓ EJERCICIOS AVANZADOS (21–30)

21. Habilitar servicio para que arranque automáticamente

sudo systemctl enable ssh

```
jcsantos at JCSantos-Ubunut in ~
└─○ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
└─○ jcsantos at JCSantos-Ubunut in ~
└─○ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-01-20 14:19:13 CET; 11min ago
     Invocation: 95cb2efaa31b49e1a47113c7aeb481ea
    TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
            man:sshd_config(5)
    Main PID: 3106 (sshd)
      Tasks: 1 (limit: 8802)
     Memory: 1.2M (peak: 2M)
        CPU: 23ms
     CGroup: /system.slice/ssh.service
            └─3106 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 20 14:19:13 JCSantos-Ubunut systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jan 20 14:19:13 JCSantos-Ubunut sshd[3106]: Server listening on 0.0.0.0 port 22.
Jan 20 14:19:13 JCSantos-Ubunut sshd[3106]: Server listening on :: port 22.
Jan 20 14:19:13 JCSantos-Ubunut systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
└─○
```

22. Deshabilitar un servicio en el arranque

sudo systemctl disable bluetooth

```
jcsantos@JCSantos-Ubunut:~
└─○ jcsantos at JCSantos-Ubunut in ~
└─○ sudo systemctl disable bluetooth.service
Synchronizing state of bluetooth.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable bluetooth
Removed '/etc/systemd/system/bluetooth.target.wants/bluetooth.service'.
Removed '/etc/systemd/system/dbus-org.bluez.service'.
└─○ jcsantos at JCSantos-Ubunut in ~
└─○
```


23. Crear un archivo de log personalizado

Ejecuta:

```
echo "Test log" | systemd-cat -t curso-linux
```

24. Buscar tu log personalizado

```
journalctl -t curso-linux
```

25. Ver logs desde el último reinicio

```
journalctl -b
```

```
journalctl -b — journalctl -b
jcsantos at JCSantos-Ubunut in ~
└─○ echo "Test log" | systemd-cat -t curso-linux
jcsantos at JCSantos-Ubunut in ~
└─○ journalctl -t curso-linux
Jan 20 14:43:32 JCSantos-Ubunut curso-linux[4235]: Test log
jcsantos at JCSantos-Ubunut in ~
└─○ journalctl -b
Jan 20 14:15:43 JCSantos-Ubunut kernel: Linux version 6.17.0-8-generic (buildd@lcy02-amd64-076) (x86_64-linux-gnu-gcc (Ubuntu 15.2.0-4ubuntu4) 15.2.0)
Jan 20 14:15:43 JCSantos-Ubunut kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.17.0-8-generic root=UUID=cfed1901-89f6-42a9-a0f3-394a5e428f79 ro deb
Jan 20 14:15:43 JCSantos-Ubunut kernel: KERNEL supported cpus:
Jan 20 14:15:43 JCSantos-Ubunut kernel: Intel GenuineIntel
Jan 20 14:15:43 JCSantos-Ubunut kernel: AMD AuthenticAMD
Jan 20 14:15:43 JCSantos-Ubunut kernel: Hygon HygonGenuine
Jan 20 14:15:43 JCSantos-Ubunut kernel: Centaur CentaurHauls
Jan 20 14:15:43 JCSantos-Ubunut kernel: zhaoxin Shanghai
Jan 20 14:15:43 JCSantos-Ubunut kernel: Disabled fast string operations
Jan 20 14:15:43 JCSantos-Ubunut kernel: BIOS-provided physical RAM map:
```

26. Identificar un proceso zombie

```
ps aux | grep Z
```

```
jcsantos@JCSantos-Ubunut:~
└─○ ps aux | grep Z
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
jcsantos  2102  0.0  0.0      0     0 ?        Z    14:15   0:00 [sh] <defunct>
jcsantos  2110  0.0  0.0      0     0 ?        Z    14:15   0:00 [snap] <defunct>
jcsantos  2112  0.0  0.0      0     0 ?        Z    14:15   0:00 [ubuntu-advantag] <defunct>
root      2141  0.0  0.0      0     0 ?        Z    14:15   0:00 [vmware-user-sui] <defunct>
jcsantos  2143  0.0  0.0      0     0 ?        Z    14:15   0:00 [ubuntu-report] <defunct>
jcsantos  4387  0.0  0.0  18148 2484 pts/0    S+   14:45   0:00 grep --color=auto --exclude-dir=.bzr --exclude-dir=CVS --exclude-dir=.git --exclude-dir=.hg --exclude-dir=.svn --exclude-dir=.idea --exclude-dir=.tox --exclude-dir=.venv --exclude-dir=venv Z
```

27. Crear un servicio systemd simple (archivo de unidad)

Crear:

```
sudo nano /etc/systemd/system/hola.service
```

Contenido:

```
[Unit]
```

```
Description=Servicio de prueba
```

```
[Service]
```

```
ExecStart=/bin/echo "Hola desde systemd"
```

```
[Install]
```

```
WantedBy=multi-user.target
```

28. Recargar systemd

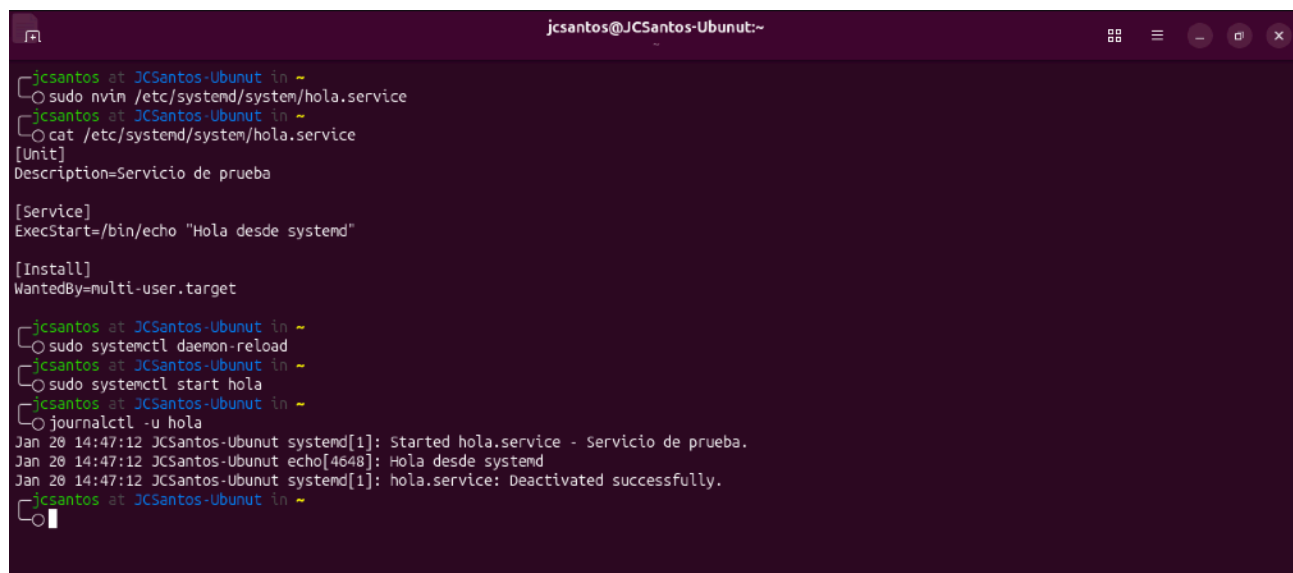
```
sudo systemctl daemon-reload
```

29. Iniciar el servicio de prueba

```
sudo systemctl start hola
```

30. Ver el log del servicio de prueba

```
journalctl -u hola
```



```
jcsantos@JCSantos-Ubuntu:~  
jcsantos at JCSantos-Ubuntu in ~  
└─○ sudo nvim /etc/systemd/system/hola.service  
jcsantos at JCSantos-Ubuntu in ~  
└─○ cat /etc/systemd/system/hola.service  
[Unit]  
Description=Servicio de prueba  
  
[Service]  
ExecStart=/bin/echo "Hola desde systemd"  
  
[Install]  
WantedBy=multi-user.target  
  
jcsantos at JCSantos-Ubuntu in ~  
└─○ sudo systemctl daemon-reload  
jcsantos at JCSantos-Ubuntu in ~  
└─○ sudo systemctl start hola  
jcsantos at JCSantos-Ubuntu in ~  
└─○ journalctl -u hola  
Jan 20 14:47:12 JCSantos-Ubuntu systemd[1]: Started hola.service - Servicio de prueba.  
Jan 20 14:47:12 JCSantos-Ubuntu echo[4648]: Hola desde systemd  
Jan 20 14:47:12 JCSantos-Ubuntu systemd[1]: hola.service: Deactivated successfully.  
jcsantos at JCSantos-Ubuntu in ~  
└─○
```

MINI-PROYECTO DEL MÓDULO 5

Objetivo:

Crear y administrar una infraestructura básica de procesos y servicios.

Tareas:

1. Crear un script en Bash llamado monitor.sh que:
 - Muestre uso de CPU
 - Muestre memoria libre
 - Muestre 5 procesos con más consumo

```
nvim ./info.sh — nvim ./info.sh
#1/bin/bash

# Colores ANSI para Ubuntu terminal
RED='\033[0;31m'
GREEN='\033[0;32m'
YELLOW='\033[1;33m'
BLUE='\033[0;34m'
PURPLE='\033[0;35m'
CYAN='\033[0;36m'
WHITE='\033[1;37m'
NC='\033[0m' # No Color

echo "CPU e Información del Procesador"
CPU=$(lscpu | grep "Model name" | sed 's/Model name: //' | xargs)
echo -e "${GREEN}Modelo:${NC} ${CYAN}$CPU${NC}"
echo -e "${GREEN}Núcleos:${NC} ${CYAN}${nproc --all}${NC}"
CPU_TOTAL=$(mpstat 1 1 | tail -1 | awk '{print $NF}' | sed 's/%//')
echo -e "${GREEN}Uso Total CPU:${NC} ${YELLOW}${CPU_TOTAL}%${NC}"
echo
# Memoria
echo "Memoria RAM"
MEM_TOTAL=$(free -h | awk 'NR==2{print $2}')
MEM_USADA=$(free -h | awk 'NR==2{print $3}')
MEM_PORC=$(free | awk 'NR==2{printf "%.1f%%", $3*100/$2 }')
echo -e "${GREEN}Total:${NC} ${CYAN}$MEM_TOTAL${NC}"
echo -e "${GREEN}Usada:${NC} ${CYAN}$MEM_USADA${NC} ${YELLOW}($MEM_PORC)${NC}"
echo
# SO y Kernel
echo "Sistema Operativo"
echo -e "${GREEN}Distribución:${NC} ${CYAN}${lsb_release -ds 2>/dev/null || cat /etc/os-release | grep PRETTY_NAME | cut -d'"'"' -f2}${NC}"
echo -e "${GREEN}Kernel:${NC} ${CYAN}${uname -r}${NC}"
echo -e "${GREEN}Hostname:${NC} ${CYAN}${hostname}${NC}"
echo -e "${GREEN}Uptime:${NC} ${CYAN}${uptime -p}${NC}"
echo
# Red
echo "Red e IP"
echo -e "${GREEN}IP Local:${NC} ${CYAN}${hostname -I | awk '{print $1}'}${NC}"
echo -e "${GREEN}Gateway:${NC} ${CYAN}${ip route | grep default | awk '{print $3}'}${NC}"
echo -e "${GREEN}MAC:${NC} ${CYAN}${ip link show | grep ether | head -1 | awk '{print $2}'}${NC}"
echo
echo "Top 5 Procesos por Consumo"
echo -e "${YELLOW}Por CPU:${NC}"
ps -eo pid,user,%cpu,%mem,cmd --sort=-%cpu | head -n 6
```

2. Crear un servicio systemd monitor.service que ejecute el script.

```
jcsantos@
jcsantos at JCSantos-Ubunut in /etc/systemd/system
cat /etc/systemd/system/monitor.service
[Unit]
Description=Servicio de Monitorizacion

[Service]
ExecStart=/etc/systemd/system/monitor.sh

[Install]
WantedBy=multi-user.target
jcsantos at JCSantos-Ubunut in /etc/systemd/system
```

3. Hacer que el servicio arranque automáticamente:

sudo systemctl enable monitor

4. Verificar funcionamiento:

systemctl status monitor

```
sudo systemctl status monitor — sudo systemctl status monitor
/etc/systemd/system
jcsantos at JCSantos-Ubunut in /etc/systemd/system
sudo systemctl enable monitor
Created symlink '/etc/systemd/system/multi-user.target.wants/monitor.service' → '/etc/systemd/system/monitor.service'.
jcsantos at JCSantos-Ubunut in /etc/systemd/system
sudo systemctl start monitor
jcsantos at JCSantos-Ubunut in /etc/systemd/system
sudo systemctl status monitor
○ monitor.service - Servicio de Monitorizacion
   Loaded: loaded (/etc/systemd/system/monitor.service; enabled; preset: enabled)
   Active: inactive (dead) since Wed 2026-01-21 09:01:50 CET; 5s ago
     Duration: 1.088s
  Invocation: 099f949e6da5467f9947071e22b8cd22
    Process: 8878 ExecStart=/etc/systemd/system/monitor.sh (code=exited, status=0/SUCCESS)
   Main PID: 8878 (code=exited, status=0/SUCCESS)
    Mem peak: 4.9M
       CPU: 114ms

Jan 21 09:01:50 JCSantos-Ubunut monitor.sh[8878]: MAC: 00:0c:29:2c:60:2e
Jan 21 09:01:50 JCSantos-Ubunut monitor.sh[8878]: Top 5 Procesos por Consumo
Jan 21 09:01:50 JCSantos-Ubunut monitor.sh[8878]: Por CPU:
Jan 21 09:01:50 JCSantos-Ubunut monitor.sh[8945]:
  PID USER      %CPU %MEM    CMD
Jan 21 09:01:50 JCSantos-Ubunut monitor.sh[8945]: 8871 jcsantos 3.6  0.4  /usr/libexec/localesearch-extractor-3 --socket-fd 3
Jan 21 09:01:50 JCSantos-Ubunut monitor.sh[8945]: 2142 jcsantos 2.4  3.8  /usr/bin/gnome-shell
Jan 21 09:01:50 JCSantos-Ubunut monitor.sh[8945]: 3050 jcsantos 1.5  3.1  /usr/bin/ptxixis --gaplication-service
Jan 21 09:01:50 JCSantos-Ubunut monitor.sh[8945]: 1991 jcsantos 0.6  0.1  /usr/bin/pipewire
Jan 21 09:01:50 JCSantos-Ubunut monitor.sh[8945]: 1 root      0.3  0.2  /usr/lib/systemd/systemd --switched-root --system --deserialize=47 auto automati
Jan 21 09:01:50 JCSantos-Ubunut systemd[1]: monitor.service: Deactivated successfully.
lines 1-20/20 (END)
```

5. Revisar logs:

`journalctl -u monitor`

```
Jan 21 09:00:58 JCSantos-Ubunut systemd[1]: Started monitor.service - Servicio de Monitorizacion.
Jan 21 09:00:58 JCSantos-Ubunut monitor.sh[8368]: CPU e Información del Procesador
Jan 21 09:00:58 JCSantos-Ubunut monitor.sh[8368]: Modelo: Intel(R) Core(TM) i9-9980HK CPU @ 2.40GHz
Jan 21 09:00:58 JCSantos-Ubunut monitor.sh[8368]: Núcleos: 128
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Uso Total CPU: 97.64%
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Memoria RAM
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Total: 7.2Gi
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Usada: 1.6Gi (21.6%)
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Sistema Operativo
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Distribución: Ubuntu 25.10
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Kernel: 6.17.0-8-generic
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Hostname: JCSantos-Ubunut
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Uptime: up 32 minutes
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Red e IP
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: IP Local: 192.168.2.132
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Gateway: 192.168.2.2
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: MAC: 00:0c:29:2c:60:2e
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Top 5 Procesos por Consumo
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8368]: Por CPU:
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8432]:      PID USER      %CPU %MEM    CMD
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8432]:      8431 root        100   0.0  ps -eo pid,user,%cpu,%mem,cmd --sort=-%cpu
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8432]:      2142 jcsantos    2.4   3.8  /usr/bin/gnome-shell
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8432]:      3050 jcsantos    1.4   3.1  /usr/bin/ptxixis --gaplication-service
Jan 21 09:00:59 JCSantos-Ubunut monitor.sh[8432]:      8266 jcsantos    1.0   0.4  /usr/libexec/localsearch-extractor-3 --socket-fd 3
lines 1-38
```

6. Documentar todo lo realizado.